

Lirui (Leroy) Wang

PERSONAL

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EDUCATION

University of Washington – Seattle, WA

B.S. Honors Computer Science, B.S. Honors Electrical Engineering

Major GPA: 3.98, 3.98

Cumulative GPA: 3.97

Expected Graduation: June 2020

Shenzhen Middle School – Shenzhen, China

Completed: June 2016

RESEARCH EXPERIENCE

Grasp Planning – UW Robotics and States Estimation Lab

Dec 2018 – Present

Apply traditional package Graspl! to a table-top object pick up tasks

Experimented with new learning-based approach that synthesize grasp pose based on shapes and geometry losses

Prototyped RL and MPC with Pybullet to map observed points directly to 6D pose increments in task space

Attempted residual policy by a neural net to imitate the MPC behavior

Jointly optimize motion and grasp planning by extending the CHOMP framework

Research extensively on traditional methods and data-driven methods in grasp synthesis

DeepIM Tracking – UW Robotics and States Estimation Lab

June 2018 – Dec 2018

Tested extensively DeepIM on YCB datasets and achieves results trained on synthetic data only

Improved network training stability, pose distribution and occlusion generation

Applied DeepIM in real-time tracking for occluded scene in YCB and AR demos

Worked on kinematics to build generalized vision system that can track robot arm

Depth and Flow – Computer Vision Course Project

March 2018 – June 2018

Investigated some deep learning methods results for depths and flow predictions

Explored unsupervised structures and ideas such as SfMlearner and MonoDepth and analyzed them on a robotic setting

Manipulation – UW Robotics and States Estimation Lab

Aug 2017 – March 2018

Solved a manipulation task with PoseCNN, which enables Baxter to manipulate detected objects

Studied camera models, robot kinematics, Baxter SDK and ROS packages for tracking and control

PUBLICATIONS AND PRESENTATIONS

RELEVANT COURSES

Computer Science:

Data Structure and Parallelism, Algorithms, Computer Vision, Introduction to Artificial Intelligence, Deep Learning, Modern Algorithm Toolbox, Graphics, Robotics, Robust Machine Learning Algorithms (Grad)

Electrical Engineering:

Digital Systems, Control System, Control Capstones, Filter Analysis, Discrete Time Linear System

Math:

Multivariable Calculus, Probability and Statistics, Numerical Analysis, Linear Analysis, Convex Optimization and Analysis (Grad), Topology, High Dimensional Statistics (Grad)

English:

Technical Writing, Advanced Technical Writing in Electrical Engineering

Statistics:

Statistical Learning (Grad)

COMPUTER SCIENCE & ENGINEERING SKILLS

Software Development Tools: Bash, Ubuntu, ROS, Eclipse, Arduino, Git, Nvidia TX Series

Programming Language Experience: Python, C++/C, Java, Matlab

Tools: Matlab, Git, Cmake, OpenCV, DJI_Guidance, Bash, Quartus, Latex, Tensorflow, Pytorch, MXNet, CUDA

Rapid Prototyping: 3D Printing, Laser Cutting, Machining, Soldering

PERSONAL DESIGN PROJECTS

SoundScape Project: Designed a sound processing pipeline from scratch for modelling the acoustics of a train whistle

uMessage: Implemented data structures such as tries and sorting algorithm such as introspective sort with HMM would drive word suggestions, spelling corrections and autocompletions.

Chess Bot: Implemented alpha-beta pruning, iterative deepening, heuristics tables, move ordering for a chess bot running in parallel.

Music Editor: Created a demo app that can record sound pieces and translate into sheet music display UI using DTFT, and it was awarded Overall 1st place out of 35 teams in Seattle

AWARDS & SCHOLARSHIPS

Dean's List, University of Washington, Fall 16 – Spring 2019.

Tau Beta Pi Scholarship, 2018

Lawrence & Lucille Frey Endowed Electrical Engineering Scholarship, 2018

Patricia G. Lynch and Theodora & Eugene Russell Memorial Computer Science Scholarship, 2019

EMPLOYMENT & LEADERSHIPS

Undergraduate Teaching Assistant, Computer Vision, Autumn 2019

Computer Vision Lead & Drone Lead, Advanced Robotics at University of Washington, 2016 - 2018

Managed a team of 10 students to design computer vision tools for our robots in a competition RoboMasters by DJI.

Perform system integration of controls, navigation and serial communication

Implement the detection, shooting, refilling, landing and communication using DJI SDK on DJI M100 drone

Intern, DJI Technology, Shenzhen, China, Jun 2017 – Aug 2017

Worked on automatic robot battle of vehicles with TX series onboard and DJI Manifold with Guidance

Implemented functions such as take-off, tag detection, and refilling to localize drone and communicate

Applied amcl to navigate and localize, and built a top-level state machine to strategize the robot

EXTRACURRICULARS

Men's Pickup Soccer Team: Throughout college years, I played regularly at pickup soccer and joined the competition organized by UW Intramural Activities Center.

Math Club: I usually joined the meetings to explore difficult problems with other students in group. This involves math problem that have less applications, but the discussion and thought process improve my problem-solving skills and meet my interests in math.

PROFESSIONAL MEMBERSHIPS

Association for Computing Machinery (ACM) Student Member

Institute of Electrical and Electronics Engineers (IEEE) Student Member